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Unit 4 - Dive into flow: Points, Lines, Surfaces, and Area of Shapes

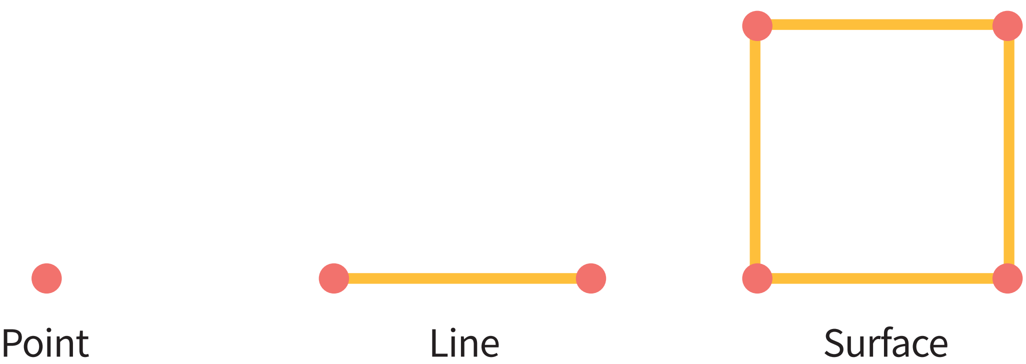
PDF Documents

PDF Document 1: 1738217749074-Chapter 4 - Unit 4 - Dive into flow Points, Lines, Surfaces, and area of shapes.pdf

This PDF document is included in the unit materials.

UNIT 3 DIVE INTO FLOW: POINTS, LINES, SURFACES, AND AREA OF SHAPES

To understand the area, we first need to learn about points, lines, and surfaces. These three concepts are fundamental to understanding space.



POINT

A point represents a single, specific location. It has no size or length, so it doesn't occupy space. For example, a tiny dot on paper marks a position without size.

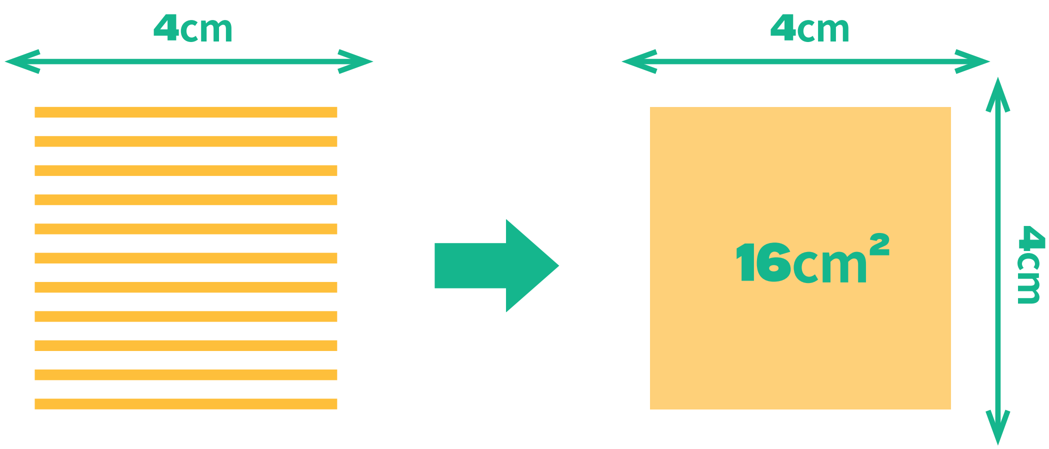
LINE

When points are arranged in a long, continuous form, they create a line. A line connects two points with length but almost no width or thickness, so it doesn’t fill space. For example, a pencil mark on paper can represent a line. Lines are used to measure length, having length but no area.

SURFACE

When lines form a flat, wide area, they create a surface. A surface has width and height, giving it a measurable size that occupies space. What we commonly call "area" is a surface's size.

HOW DOES IT WORK

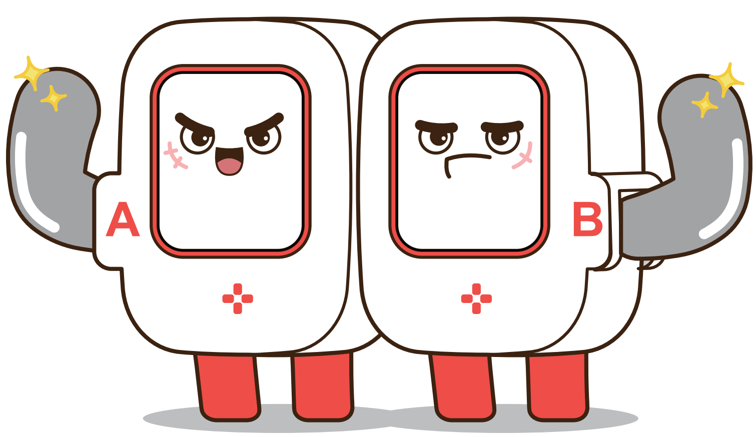


Looking at the illustration, we can see how the area is calculated by stacking multiple lines.

First, we know the length of one line, which is 4 cm. By stacking these lines, we create a square with dimensions of 4 cm × 4 cm.

The area of this square can be calculated by multiplying the length and width: 4cm×4cm=16cm2

So, the surface area created by stacking lines is 16 square centimeters.



Multiple Choice Questions

1 What is the correct difference between analog and digital?

Ⓐ Analog is a broken form, while digital is smooth.

Ⓑ Analog is continuous and smooth, while digital is expressed in discrete numbers.

Ⓒ Electrical signals represent analog, while digital signals are only represented by sound.

Ⓓ Analog is only used in electronic devices.

CORRECT ANSWER Ⓑ

2 Which of the following is an advantage of a digital ruler?

Ⓐ It is lightweight.

Ⓑ It can measure length more precisely.

Ⓒ It represents continuous length.

Ⓓ It has a variety of shapes.

CORRECT ANSWER Ⓑ

3 Among the units of length, approximately how many centimeters is 1 inch?

Ⓐ About 1.5 cm

Ⓑ About 2 cm

Ⓒ About 2.54 cm

Ⓓ About 3 cm

CORRECT ANSWER Ⓒ

Reflection Questions

1 Why do you think there’ s a difference between analog and digital?

2 How could a digital ruler be useful in everyday life? Can you give an example?

3 What is one limitation of the digital ruler we created? How could it be improved?